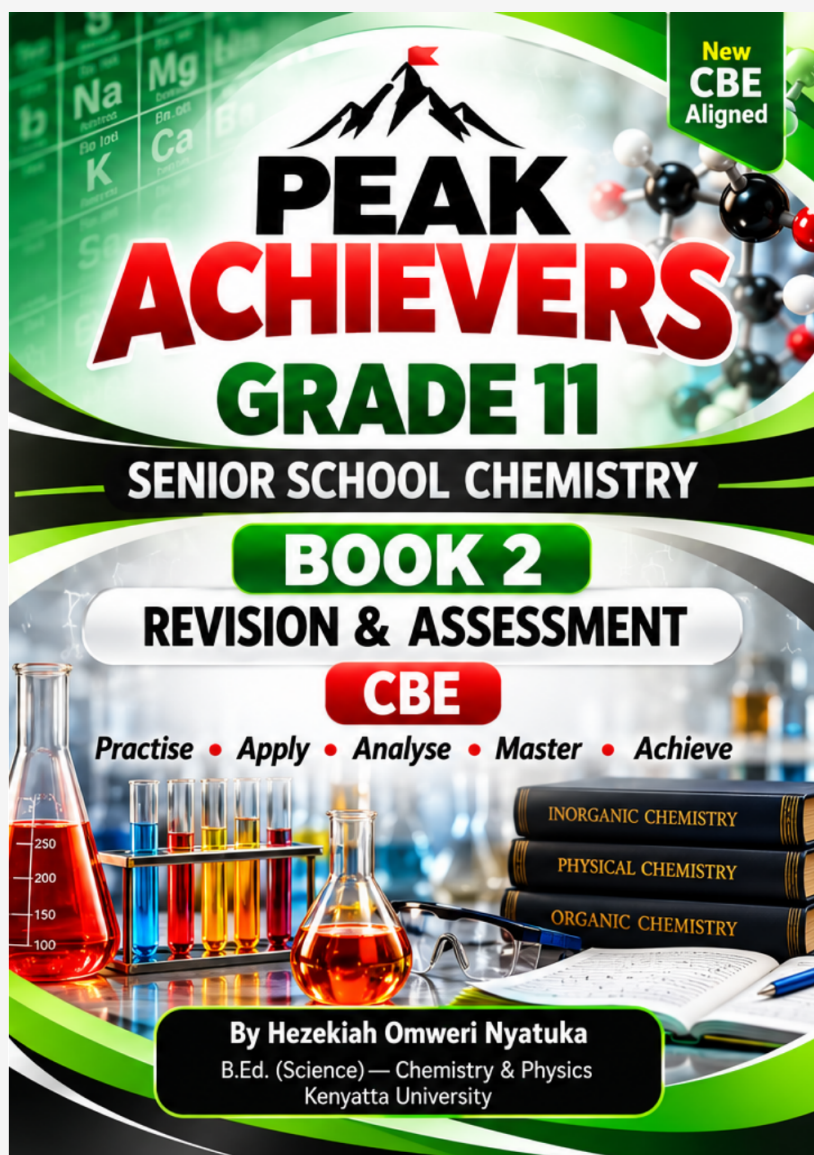


PEAK ACHIEVERS

GRADE 11 SENIOR SCHOOL CHEMISTRY

FREE SAMPLE PACK - BOOK 2

REVISION & ASSESSMENT



FULL BOOK PRICE
KSh 550

This sample includes:

- selected notes/pages
- worked examples & questions
- practical/application content
- a preview of assessment style

Website

www.peakachiever.co.ke

WhatsApp / Enquiries

0740166105 / 0740355410

FREE PREVIEW ONLY - NOT FOR RESALE OR REDISTRIBUTION

Selected pages from the complete Grade 11 Chemistry Book 2.

PEAK ACHIEVERS
GRADE 11
SENIOR SCHOOL CHEMISTRY
BOOK 2
REVISION & ASSESSMENT
CBE

Practise • Apply • Analyse • Master • Achieve

By Hezekiah Omweri Nyatuka

B.Ed. (Science) — Chemistry & Physics
Kenyatta University

Copyright

Copyright © 2026 Hezekiah Omweri Nyatuka / Peak Achievers. All rights reserved.

Peak Achievers Grade 11 Senior School Chemistry — Book 2: Revision & Assessment is an independently developed supplementary learning resource intended to support Grade 11 Chemistry within Kenya's Competency Based Education framework.

It is not an official publication of KNEC or KICD, and the practice assessments in this book should not be interpreted as official examination papers.

First Edition: 2026

Dedication

This book is dedicated to the loving memory of my late father, Simeon Nyatuka, whose life, guidance and sacrifices remain a lasting source of inspiration.

To my mother, Hyrine Kerubo; my guardians, Peter Munde Mong'are and Teresa Kwamboka Sawe; my friends Mnaree, Richie, Duke, Caleb, Cuba, Philo, Rob, Tim, Mr Peter Kimani and many others; to Her; to my fellow teachers; and to every learner determined to improve.

May every question you attempt take you one step closer to becoming the achiever you are capable of being.

— Hezekiah Omweri Nyatuka

Preface

Chemistry is mastered not only by reading concepts but also by applying them, solving problems, interpreting evidence and learning from mistakes. Book 2 is the revision and assessment companion to Book 1: Concise Notes.

It covers The Mole, Non-metals, Salts, Gas Laws, Reaction Rates and Hydrocarbons through short revision notes, topical quizzes, mixed-concept quizzes, practical and inquiry tasks, calculations, real-life applications and final practice assessments.

How to Use This Book

REVISE → ATTEMPT → APPLY → CHECK → CORRECT → MASTER

1. Read the Quick Revision Notes at the start of each topic.
2. Attempt the Topical Quiz before checking answers.
3. After all six topics, attempt the five Mixed Concept Quizzes.
4. Attempt the six Final National Assessment Practice Tests under timed conditions where possible.
5. Use the Answers & Marking Guidance to diagnose errors, not merely to count marks.
6. Retry difficult questions after correcting your work.

Peak Achievers Feature Key

-  REMEMBER — important fact or relationship.
-  HINT — guidance without giving away the full solution.
-  COMMON ERROR — a frequent mistake.
-  PRACTICAL INQUIRY — laboratory and investigation skills.
-  APPLICATION — Chemistry in everyday life, industry or the environment.
-  THINK BIG — deeper reasoning and integration.
-  EXAM FOCUS — assessment technique or key concept.
-  PEAK CHALLENGE — demanding application problem.

Contents

PART I — Topical Revision & Quizzes	3–17
Topics: Mole 4 • Non-metals 6 • Salts 8 • Gas Laws 10 • Reaction Rates 13 • Hydrocarbons 16	
PART II — Mixed Concept Quizzes 1–5	18–23
PART III — Final National Assessment Practice Tests 1–6	24–32
PART IV — Answers & Marking Guidance	33–46
Closing Page	47

PART I

TOPICAL REVISION & QUIZZES

Revise one concept at a time. Build confidence.

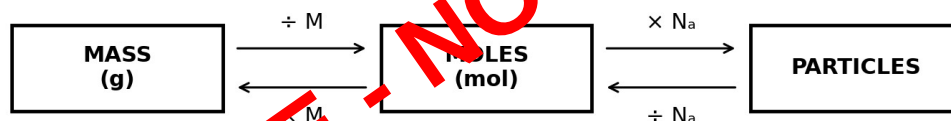
FREE SAMPLE - NOT FOR RESALE

TOPIC 1 — THE MOLE

Quick Revision Notes

- 1 mole contains 6.022×10^{23} specified particles.
- $n = m/M$; $N = nN_a$; $N_a = 6.022 \times 10^{23} \text{ mol}^{-1}$.
- Use balanced-equation coefficients as mole ratios.
- Percentage composition = (mass of element / formula mass) $\times 100\%$.
- Empirical formula: convert masses/percentages to moles, divide by the smallest, obtain simplest whole-number ratio.
- Molecular formula = empirical formula $\times (M_r / \text{empirical-formula mass})$.
- Concentration: $C = n/V$, with V in dm^3 .
- Standard solution route: WEIGH $\rightarrow$ DISSOLVE $\rightarrow$ TRANSFER $\rightarrow$ RINSE $\rightarrow$ MAKE TO MARK $\rightarrow$ MIX.

Mole Conversion Map



Balanced equation coefficients connect moles of different substances.

Figure 1.1 Mole conversion map.

Preparing a Standard Solution

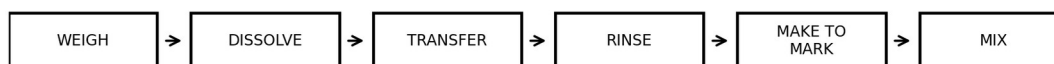


Figure 1.2 Standard-solution preparation sequence.

Topical Quiz

1. Define one mole. [1 mark]
2. State Avogadro's constant. [1 mark]
3. Calculate M_r of H_2SO_4 . [2 marks]
4. Calculate the amount in moles in 8.0 g of O_2 . [2 marks]

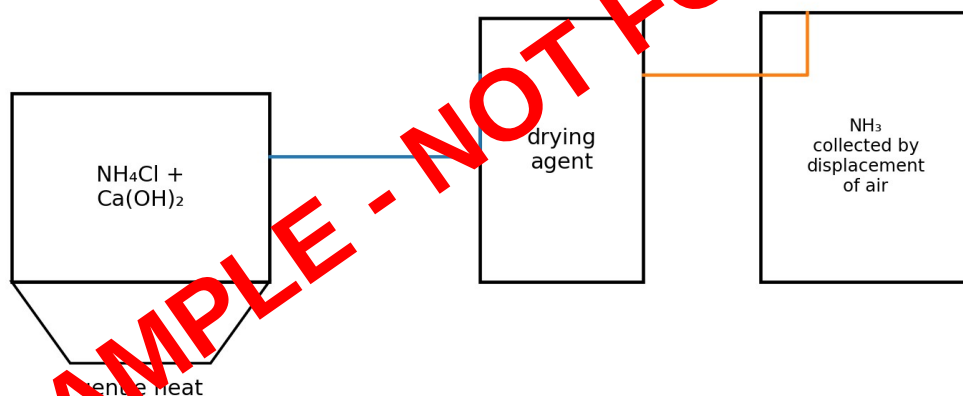
5. Calculate the number of molecules in 0.25 mol of CO_2 . [2 marks]
6. Magnesium burns in oxygen. Calculate the mass of MgO formed from 12 g Mg . [4 marks]
7. $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$. Calculate the mass of CaO formed from 25 g CaCO_3 . [4 marks]
8. A compound contains 24 g carbon and 4 g hydrogen. Determine its empirical formula. [4 marks]
9. A compound has empirical formula CH_2 and $M_r = 56$. Determine its molecular formula. [3 marks]
10. Calculate the concentration of 0.20 mol solute in 500 cm^3 solution. [3 marks]
11. Describe how to prepare a standard solution using a volumetric flask. [5 marks]
12. Explain one industrial use of mole calculations in reducing waste/cost. [4 marks]
13. Explain why identifying the limiting reactant matters when predicting product formed. [4 marks]
14. A 50 g limestone sample is 80% CaCO_3 . Calculate the theoretical mass of CaO formed on complete decomposition. [5 marks]

TOPIC 2 — NON-METALS

Quick Revision Notes

- Nitrogen forms about 78% of air and is relatively unreactive because of the strong $\text{N}\equiv\text{N}$ bond.
- Ammonia is colourless, pungent, highly soluble in water and alkaline; damp red litmus turns blue.
- Laboratory preparation: $2\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCl}_2 + 2\text{NH}_3 + 2\text{H}_2\text{O}$.
- Haber process: $\text{N}_2 + 3\text{H}_2 \rightleftharpoons 2\text{NH}_3$; iron catalyst; unreacted gases are recycled.
- Sulfur burns: $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$. Conversion to SO_3 uses $2\text{SO}_2 + \text{O}_2 \rightleftharpoons 2\text{SO}_3$ with V_2O_5 catalyst.
- Chlorine in water: $\text{Cl}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCl} + \text{HClO}$.
- Chloride test gives a white AgCl precipitate: $\text{Ag}^+ + \text{Cl}^- \rightarrow \text{AgCl}(\text{s})$.
- Excess fertiliser runoff can enrich water bodies and contribute to oxygen depletion.

Laboratory Preparation and Collection of Ammonia



Ammonia is highly soluble in water, so it is not collected over water.

Figure 2.1 Simplified laboratory preparation of ammonia.

Simplified Haber Process



Figure 2.2 Simplified Haber-process flow and recycling.

15. Explain why a smoky yellow flame indicates poor combustion. [2 marks]
16. Two gases X and Y are ethane and ethene. Bromine water is decolourised by Y only. Identify X and Y and explain. [3 marks]
17. THINK BIG: C_4H_{10} and C_4H_8 are compared. Classify each within the series studied and state one chemical test that supports the distinction. [4 marks]
18. PEAK CHALLENGE: A household stove produces soot and a yellow flame. Explain the chemistry and give two actions that reduce the hazard. [4 marks]

FREE SAMPLE - NOT FOR RESALE

PART II

MIXED CONCEPT QUIZZES

Choose the chemistry. Connect the concepts.

FREE SAMPLE - NOT FOR RESALE

MIXED CONCEPT QUIZ 1

Suggested Time: 45 minutes

1. State Avogadro's constant. [1 mark]
2. Explain why ammonia is not collected over water. [2 marks]
3. Write the general formula of the alkenes studied. [1 mark]
4. Name the gas that turns limewater milky. [1 mark]
5. Convert 47°C to kelvin. [1 mark]
6. State two factors that can increase reaction rate. [2 marks]
7. State the observation when ethene is shaken with bromine water. [1 mark]
8. Write the ionic equation for formation of silver chloride. [2 marks]
9. Calculate the amount in 22 g CO_2 . [2 marks]
10. Calculate the mass of MgO formed from 12 g Mg . [3 marks]
11. A gas occupies 600 cm^3 at 100 kPa. Find its volume at 150 kPa at constant temperature. [3 marks]
12. Calculate the concentration of 0.15 mol solute in 300 cm^3 solution. [3 marks]
13. Design a fair investigation of the effect of CaCO_3 particle size on rate with HCl . [5 marks]
14. Describe how to prepare pure dry CuSO_4 crystals from CuO and dilute H_2SO_4 . [5 marks]
15. Explain how fertiliser runoff can reduce dissolved oxygen in water. [3 marks]
16. Explain one danger of incomplete charcoal combustion indoors. [2 marks]
17. THINK BIG: Explain how accurate mole ratios and catalysts can reduce industrial waste and energy use. [4 marks]

MIXED CONCEPT QUIZ 5

Suggested Time: 45 minutes

1. Calculate the amount in 18 g H_2O . [1 mark]
2. Name the catalyst used in conversion of SO_2 to SO_3 . [1 mark]
3. State one method for preparing a soluble salt and one for an insoluble salt. [2 marks]
4. State Boyle's law. [2 marks]
5. Explain the effect of temperature on particle kinetic energy. [2 marks]
6. State the general formula of alkanes. [1 mark]
7. State the observation in a positive chloride-ion test. [1 mark]
8. Calculate the number of molecules in 0.50 mol CO_2 . [2 marks]
9. A sample contains 20 g MgO . Calculate the mass of Mg that reacted, assuming complete formation of MgO . [3 marks]
10. A gas occupies 240 cm^3 at 300 K. Calculate its volume at 375 K at constant pressure. [3 marks]
11. Calculate the concentration of 0.10 mol NaCl in 250 cm^3 solution. [3 marks]
12. Describe preparation of pure dry CuSO_4 crystals from CuO and dilute H_2SO_4 . [5 marks]
13. A reaction produces 60 cm^3 gas in 30 s. Calculate average rate and explain how a catalyst changes the rate. [3 marks]
14. C_5H_{10} fits the general formula of which hydrocarbon series studied? State a chemical test that supports identification as an alkene. [3 marks]
15. Write the equation for sulfur burning in oxygen and state one environmental concern. [2 marks]
16. Equal moles of CaCO_3 powder and chips react separately with excess acid. Predict the relative rates and final CO_2 volumes. Explain. [3 marks]
17. A gas occupies 2.0 dm^3 at 100 kPa and 300 K. Calculate its volume at 120 kPa and 360 K. [3 marks]
18. Give one example showing how Chemistry can improve safety or resource use in everyday life. [3 marks]

FINAL NATIONAL ASSESSMENT PRACTICE TESTS

Apply knowledge • Interpret evidence • Solve problems

FREE SAMPLE - NOT FOR RESALE

READY FOR THE COMPLETE BOOK?

PEAK ACHIEVERS - GRADE 11 CHEMISTRY

BOOK 2 - KSh 550

Topical Quizzes • Mixed Concepts • Final Practice Tests • Answers

BOOK 1

KSh 450

BOOK 2

KSh 550

BOTH BOOKS

KSh 650

BEST VALUE - SAVE KSh 350

Order / Learn More

www.peakachiever.co.ke

WhatsApp / Call: 0740166105 / 0740355410

Email: hezekiahomweri@gmail.com

Copyright © 2026 Hezekiah Omweri Nyatuka / Peak Achievers

This free sample is provided for personal educational preview only.

Redistribution, resale or unauthorised reproduction of the complete books is prohibited.